

COGNITIVE VULNERABILITIES IN HEALTHCARE SYSTEMS: HOW SAFE ARE PATIENTS?

MAPPING BETWEEN THE FLAGS SYSTEM TO SECURITY CONCEPTS – HUMAN-LAYER VULNERABILITIES

THREAT MODELLING

THE ASSET	THE PATIENT
THE ADVERSARY	1: DISEASE 2: INJURY 3: HUMAN ERROR (WORKLOAD, CULTURE, SYSTEM PRESSURE)
RISKS	1: WORSENING STATE 2: COMPLICATIONS 3: DEATH
COMPLICATIONS OF DISEASE / INJURY = MORE SOPHISTICATED ATTACKS	

COMMON PATIENT COGNITIVE VULNERABILITIES	
1. PERSISTENT HOT STATE	2. TOO MUCH TRUST
3. TOO MUCH EMPATHY FOR NURSES/DOCTORS	4. WHITE COAT SYNDROME
5. FEAR OF "WASTING" OTHERS TIME/BEING BOTHERSOME	

CHART KEY COLOURS

TYPE 1 ERROR	TYPE 2 ERROR
SIDE CHANNEL	INSIDER RISK

HOW TO IMMEDIATELY ASSESS YOUR OWN SAFETY

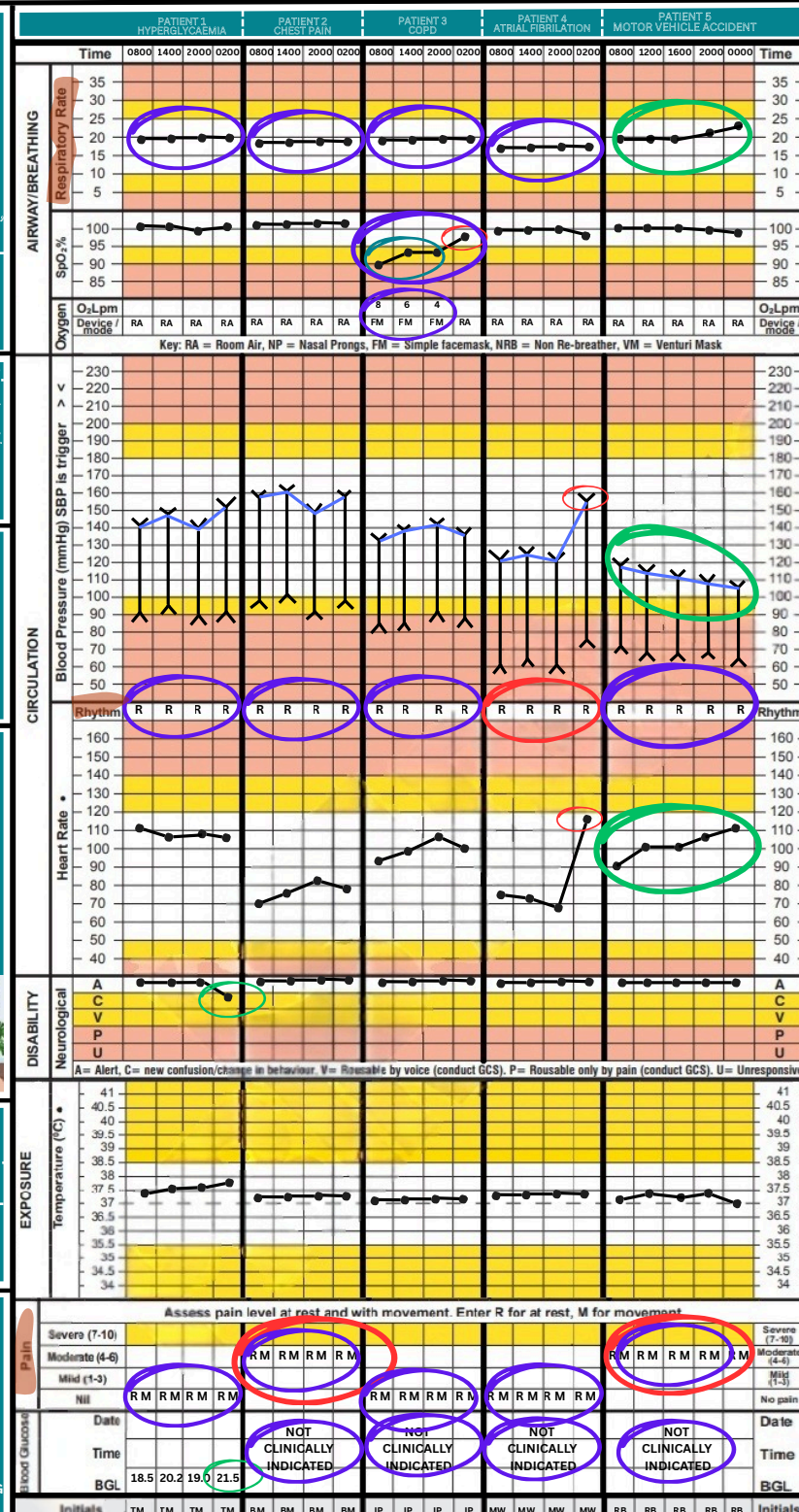
IF EVER IN HOSPITAL, DURING BLOOD PRESSURE TIME THE PRACTITIONER DOES NOT FEEL PULSE OR COUNT YOUR RESPIRATORY RATE (CHEST RISE AND FALL) THIS MAY INDICATE A PROCESS GAP



C	PACCESS TO PATIENT DATA AND PRACTITIONER ACTIONS
I	ALL PATIENTS DISPLAYED HAVE TAMPERED DOCUMENTATION
A	INACCURATE RECORDS LIMIT ACCESS TO VERIFIED PATIENT INFORMATION

DOCUMENTATION ACCURACY CAN DEGRADE UNDER WORKLOAD, TIME PRESSURE, AND DEFENSIVE PRACTICE

FOR EXAMPLE, RESPIRATORY RATE IS OFTEN INCONSISTENTLY RECORDED AS NON-VENTILATED PATIENTS RELY ON MANUAL MEASUREMENT, CREATING RISK DESPITE BEING AN EARLY INDICATOR OF DETERIORATION



THEORETICAL FAILURE SCENARIOS

- MISSING SIDE CHANNEL INFORMATION (BGL 21.5 - YELLOW ZONE)
SYSTEM DOES NOT DISPLAY THIS + COMBINED WITH NEUROLOGICAL CHANGE = RED ZONE ESCALATION CRITERIA (DUE TO 2 YELLOWS)
- INACCURATE DOCUMENTATION UNDER WORKLOAD PRESSURE
THIS PATIENT CAME IN WITH CHEST PAIN (THIS IS A PRE-HEART ATTACK SYMPTOM - THE PAIN IS EXTREME). DOCUMENTED SCORE MAY UNDERESTIMATE SEVERITY, AFFECTING ESCALATION
- OVER RELIANCE ON PROTOCOLS WITH LIMITED CLINICAL CONTEXT
COPD HAS A NORMAL RANGE OF 88-92 SPO2%, SUPPLYING OXYGEN TURNED THIS TYPE 1 ERROR INTO A TYPE 2
- DELAYED OR MISSED ESCALATION
AF HAS AN IRREGULAR HEART RHYTHM. THE TYPE 2 ERRORS SHOWN ARE SHARP INCREASE FROM BASELINE FOR HEART RATE AND BLOOD PRESSURE. ESCALATION WAS DELAYED OR NOT INITIATED
- FAILED COMPENSATION DETECTION (SYSTEM + HUMAN FACTORS)
THE SIDE CHANNELS ARE CLUSTERED TYPE 2 ERRORS. IN SHORT AS BLOOD PRESSURE GOES DOWN HEART RATE AND RESP RATE COMPENSATE BY INCREASING. WHEN COMPENSATION FAILS THIS PATIENT WILL GO INTO SHOCK WITH A 30% SURVIVAL RATE

OTHER COMMON INSIDER COGNITIVE VULNERABILITIES

- PERSONAL BIASES INFLUENCING DECISIONS
- AVOIDANCE OF HIGH-DEMAND INTERACTIONS

SOME WAYS TO PROTECT YOURSELF!

- TAKE YOUR OWN VITAL SIGNS WITH TIME STAMPS
- REMEMBER YOU ARE THE CENTER OF CARE
- DO NOT HESITATE TO ASK FOR HELP. STAFF ARE THERE TO SUPPORT YOUR CARE
- IF YOU HAVE SAFETY CONCERNS, USE THE REACH SERVICE OR HCCC
<https://lhd.health.nsw.gov.au/reach#how-to-call-reach>
<https://www.hccc.nsw.gov.au>

CARRIEDS1